

GURAHONT, Romania

WMO: 151820

Lat: 46.2792N

Lon: 22.3333E

Elev: 178

StdP: 99.20

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-8.3	-13.2	1.2	-8.7	-10.9	1.5	-7.2	6.8	1.6	5.3	2.8	1.0	120	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	33.2	21.1	31.3	20.9	29.5	20.4	22.7	29.6	21.9	28.9	21.2	27.9	1.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.7	24.9	19.9	14.9	24.1	19.1	14.2	23.5	67.7	29.5	64.9	29.1	62.2	27.7	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.0	4.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-0.4	2.8	5.9	11.1	16.3	20.0	21.0	21.9	16.3	11.6	7.2	1.9
	DBStd	8.48	4.60	4.34	4.18	4.54	3.04	3.25	2.95	3.21	4.08	4.23	4.58	3.90
	HDD10.0	1101	323	204	138	41	1	0	0	0	2	34	108	250
	HDD18.3	2873	582	436	386	217	76	19	10	8	87	211	334	508
	CDD10.0	1589	0	2	11	75	198	299	339	368	191	82	24	0
	CDD18.3	321	0	0	0	1	14	69	92	117	26	1	0	0
	CDH23.3	3454	0	0	1	52	193	697	944	1236	290	42	0	0
	CDH26.7	1271	0	0	0	5	29	228	356	558	93	2	0	0
Wind	WSAvg	1.5	1.2	1.5	1.7	1.8	1.6	1.5	1.6	1.5	1.4	1.4	1.4	1.3
Precipitation	PrecAvg	835	58	44	56	66	81	97	98	64	76	67	63	66
	PrecMax	1098	94	127	164	118	150	181	178	133	183	141	127	145
	PrecMin	579	22	10	14	17	39	30	34	12	14	4	5	9
	PrecStd	134	21	27	36	27	28	36	45	32	38	37	26	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	16.4	21.7	27.1	29.2	32.7	34.5	35.9	32.4	26.6	21.0	13.6
		MCWB	9.2	10.2	12.9	16.1	18.3	22.2	21.3	21.5	19.6	16.4	13.6	9.3
	2%	DB	9.1	13.9	18.5	24.7	27.3	30.9	32.7	34.1	28.9	24.3	18.9	10.8
		MCWB	6.7	8.6	11.0	14.8	18.1	21.6	21.2	20.7	18.5	15.7	12.9	8.1
	5%	DB	7.2	11.4	15.5	22.3	25.6	29.3	30.5	31.7	26.7	21.8	16.4	8.9
		MCWB	5.5	7.6	9.5	13.9	17.2	20.8	20.8	20.8	18.2	15.0	11.9	6.8
	10%	DB	5.4	9.3	12.9	19.7	23.7	27.4	28.3	29.7	23.9	18.7	14.0	7.0
		MCWB	4.0	6.4	8.0	12.6	16.4	20.2	20.2	20.7	17.5	13.8	10.8	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	11.2	13.4	17.1	20.6	24.1	23.2	23.9	20.8	17.8	14.4	10.1
		MCDB	12.8	15.4	21.5	24.9	27.0	30.1	29.6	30.6	30.1	23.0	19.4	12.9
	2%	WB	7.3	9.4	11.5	15.8	19.1	22.7	22.3	22.6	19.8	16.5	13.4	8.2
		MCDB	8.7	12.8	17.6	23.2	24.9	29.0	29.3	30.1	26.3	22.4	17.8	10.6
	5%	WB	5.7	8.1	10.0	14.6	18.1	21.6	21.6	21.9	18.9	15.6	12.5	6.9
		MCDB	7.0	11.1	14.3	21.4	23.7	27.8	28.4	29.1	25.0	20.9	16.4	9.0
	10%	WB	4.0	6.5	8.7	13.4	17.1	20.6	20.9	21.1	18.0	14.4	11.1	5.6
		MCDB	5.3	9.0	12.4	19.1	22.2	26.4	26.9	28.2	22.7	18.3	13.8	6.9
Mean Daily Temperature Range	5% DB	MDBR	6.5	8.6	10.5	11.8	11.7	11.9	12.7	13.3	11.6	11.4	8.8	6.0
		MCDBR	8.6	11.9	14.6	16.1	15.2	14.5	16.2	16.8	15.8	16.6	12.2	8.9
		MCWBR	6.2	7.4	8.1	7.7	7.0	6.1	6.0	5.7	6.0	8.2	7.0	6.1
	5% WB	MCDBR	8.3	10.9	13.3	14.6	13.9	13.7	13.7	14.8	13.5	14.8	11.1	8.9
		MCWBR	6.0	6.8	7.7	7.0	6.8	6.0	5.7	5.8	5.5	7.4	6.3	6.1
Clear-Sky Solar Irradiance	taub	0.327	0.355	0.397	0.452	0.438	0.432	0.449	0.455	0.421	0.381	0.352	0.328	
	taud	2.337	2.271	2.209	2.078	2.180	2.235	2.195	2.176	2.239	2.342	2.373	2.370	
	Ebn at Noon	755	803	819	804	831	837	816	793	786	765	718	709	
	Edh at Noon	77	99	122	152	143	136	139	137	118	92	74	67	
All-Sky Solar Radiation	RadAvg	1.21	2.02	3.18	4.38	5.28	5.84	5.73	5.26	3.73	2.56	1.48	0.96	
	RadStd	0.16	0.27	0.45	0.45	0.50	0.50	0.49	0.51	0.41	0.30	0.19	0.13	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (67 neighbors)		+0.71	N/A	N/A	+0.97	N/A	N/A	-127	-193	+117	+58	

Nomenclature: See separate page